



LBS

LEVEL BUILDING SIDEKICK

User Manual 2025

2025 Version 0.2.0

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LBS - Tool Overview

LBS is a mixed-initiative content generation tool focused on creating content such as levels, population and quests.

Level Building Sidekick (LBS) allows interactive software developers to define the architecture, constraints, graphical elements and decorative details to be used in the construction of a proposed game level. This process allows an increase of the quality and diversity of content they can offer their players. The tool allows artists to upload any type of art they create for their game, allowing them to compose custom art sets for different genres, sections or styles.

Assistant

The intelligent Assistant automatically suggests options or variations of this content that share common characteristics with what was initially established by the developers. Thus, when creating a level proposal, it is easy to have a variety of alternative configurations thanks to the different artistic resources that can be loaded by the designer and the suggestions given by the intelligent assistants.

Unity integration

Level Building Sidekick runs as an extension of the Unity game engine. This tool adds a series of new customizable windows within the development environment, following the philosophy of the engine to work all disciplines related to it within the same and not having the need to use multiple tools at once. LBS also allows users to save the modifications in a serialized way, so that they can be shared with other developers and they can continue modifying.

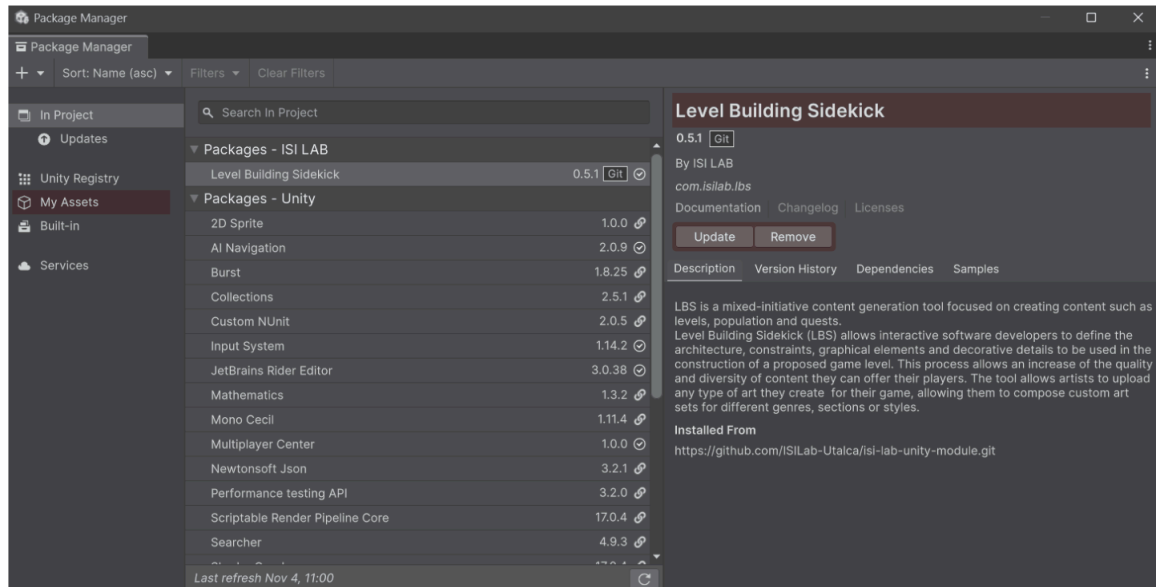
The architecture of this tool has an element composition orientation, allowing LBS to continue to be extended by both the development team and its users. In addition, this approach makes LBS a great research platform, as it allows new AI techniques to be added on top of the same tool without having to start from scratch, cutting development times and allowing the researcher to focus on the relevant areas of their research.

LBS has four main features:

- Interior space design.
- Exterior space design using wave function collapse.
- Game asset placement.
- Mission design.

Installing LBS | Package Setup

Setup



LBS, an extension for Unity Engine, follows a standard installation process similar to other Unity extensions. Below are the necessary steps for installation.

First, check if we have Level Building Sidekick in our Assets looking at the package manager: Windows > Package Manager > My Assets

Select the **Level Building Sidekick** Package and click on Install button.

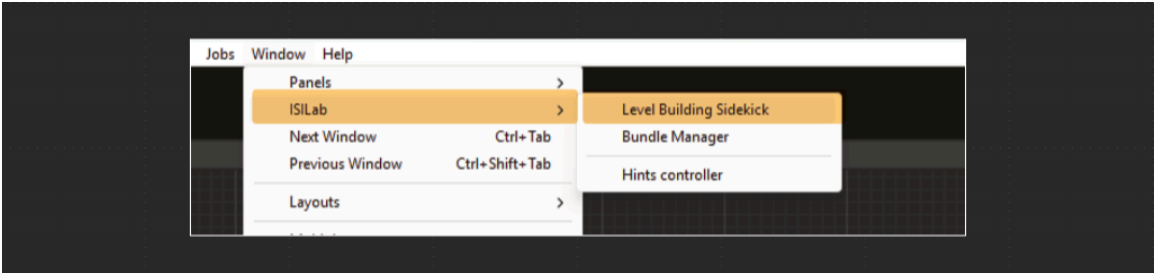
The indicated packages must be imported for LBS to function properly. Clicking on **Install/Upgrade** will automatically install the required packages.

Requirements

This extension is developed to be compatible with **Unity 6 versions 6000.0.59f2** or higher.

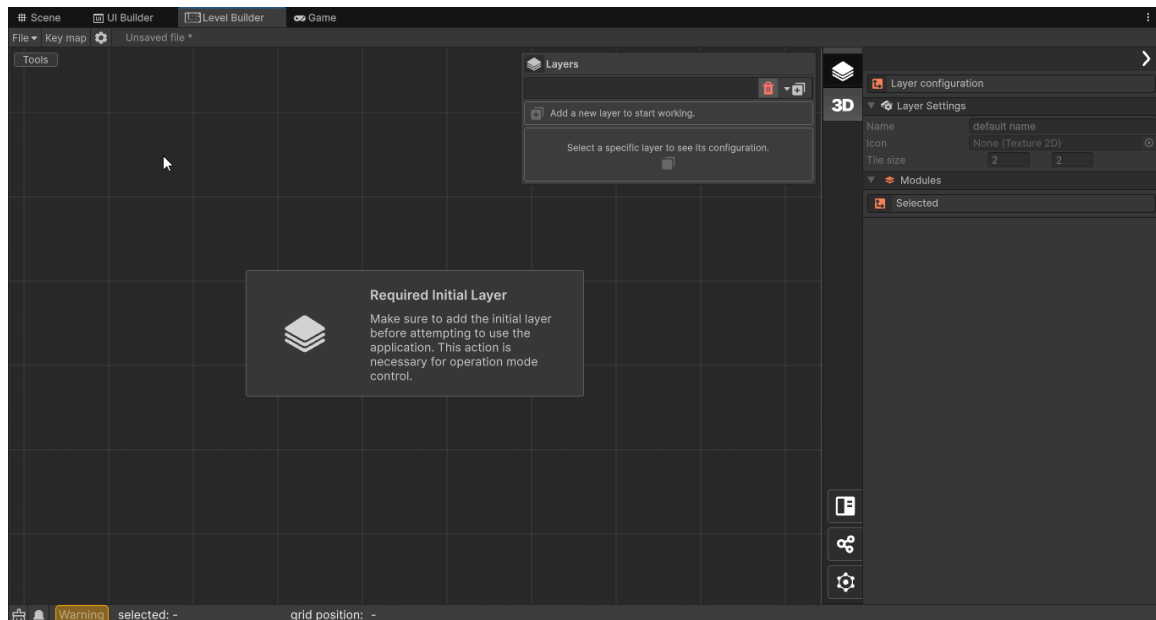
The installation requires some dependencies packages like newtonsoft.JSON, VectorGraphics . The engine will **automatically** install the dependencies when installing the LBS package and open a window that will facilitate the importation process.

To start working click on the **Window** tab, go to the **ISILab** section and then click on **Level Building Sidekick**, to bring up the tool's working window.



Tool Overview

Main Window

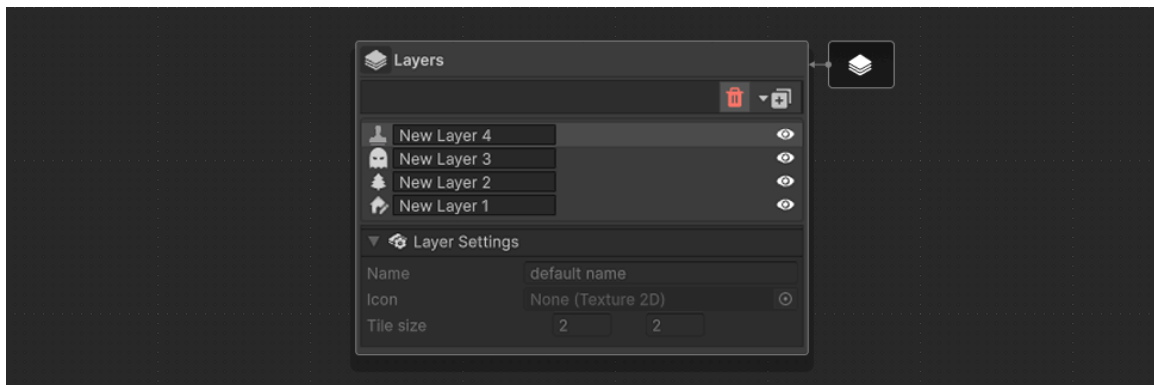


https://isilab-utalca.github.io/isilab-website/documentation/tutorials/lbs/getting_started/Unity_1OnOncbeB3.png

In the main window, you will find a large working area visualized with a tiling of squares, in this area you can place, move, and remove elements to shape your levels. To the right of this, you will find an internal inspector that will allow you to work with the specific values of the elements to be modified, their base behaviors, and the assistants that will help you work with this tool.

In addition to this, we can find 2 general buttons in the top of the vertical bar between these two aforementioned elements , which will allow showing and hiding 2 main panels to organize the work. The first corresponds to a panel to organize the content of the level by layers, the second corresponds to the panel that will allow us to pass these layers to structures usable in unity directly .

Layers



https://isilab-utalca.github.io/isilab-website/documentation/tutorials/lbs/getting_started/layer_info_01.png

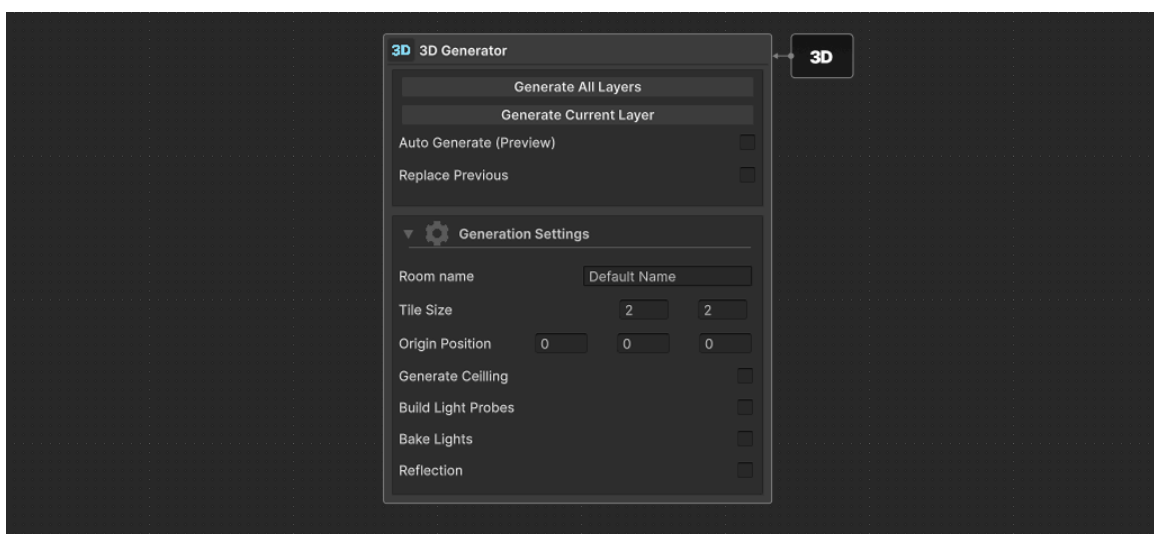
The layers panel can be displayed by toggle the **Layers** button in the side panel, you can press it again to hide the panel.

The layers that are created are displayed here. For each type of layer you will have different working tools, and the inspector view will change as well.

There are currently 3 types of layers:

- Map generation layer. See [Module-1A \(Interior space design\)](#) or [Module-1B \(Exterior space design\)](#)
- Entities population layer. See [Module-2](#)
- Quest design layer.

3D Generation



https://isilab-utalca.github.io/isilab-website/documentation/tutorials/lbs/getting_started/3DInfo_01.png

The 3D generator panel can be displayed by toggle the **3D** button in the side panel, you can press it again to hide the panel.

Features

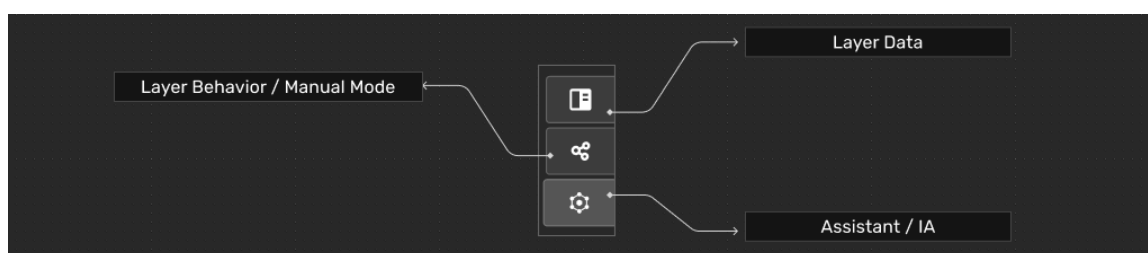
When generating you can choose to build a single layer (the selected one) or all the layers in the current project. The generator can be tweak with some configuration option:

- **Replace Previous:** Allows replace of same name layer(s).
- **Tile Size:** override the default tile size of the set.

Features for Interior space design (step 1A)

- **Generate Ceiling:** Close the room and use the **Ceiling** bundle.
- **Build Light Probe:** Generate light probes inside each room, allow easy dynamic lighting.
- **Bake Lights:** Automatically bake light for high quality static lighting.
- **Build Reflections:** Generate a **ReflectionProbe** Object in the center of each room.

LBS Inspector Panel



https://isilab-utalca.github.io/isilab-website/documentation/tutorials/lbs/getting_started/Step_1A_Info_Side_panel_01.png

From the bottom-left side panel you can access to a variety of layer specific sections:

- **Layer data:** The information of the selected object in the workspace is displayed.
- **Layer Behavior / Manual Mode:** You can access brushes and other options for working manually in the workspace.
- **Assistants / AI:** Here you will find AI assistance with different options and features depending on the layer's type.

LBS - Bundles

What are Bundles?

“Bundles” are intermediary scriptable objects that wrap different game assets, models, prefabs and its metadata used by LBS to generate interior, exterior, and population layer content. A bundle can contain one or more prefabs inside. A bundle can also contain **characteristics**, characteristics are the base metadata that LBS uses to know how to place bundles (and their assets) in the **3D generation process**.

The dimensions and settings of each prefab will depend exclusively on the users at the moment of using the 3D generation tab we recommend check the transform of each prefab and secure its in {position: }

Main Bundles

Main Bundles are scriptable objects that contain a list of bundles. It is recommended to use Main Bundles for same-layer-bundles. You may also separate same-layer-bundles in different Main Bundles depending on their purpose. Example: A project with 4 main bundles, in which two of them are population layers, one for props and one for lighting assets; InteriorLayer_MB, ExteriorLayer_MB, PopulationAssets_MB and PopulationLighting_MB.

Types of Bundles

Here are some of the considerations to keep in mind when creating Bundle's Prefabs depending on their destined layer.

Interior Layer Bundles

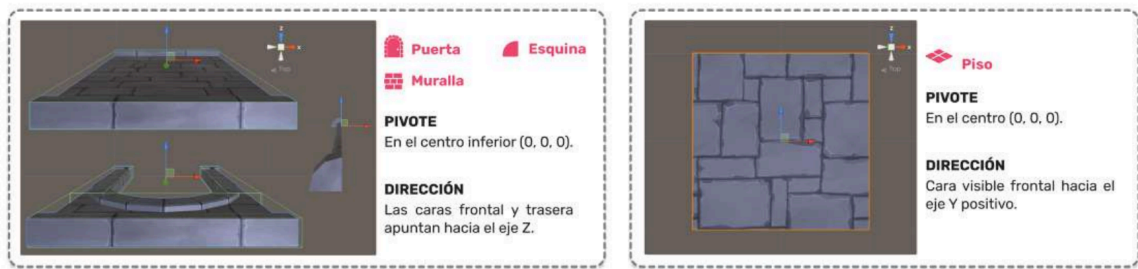
The elements of this layer consist mainly of floor, doors, walls and corners. Prefab's considerations:

- Its position is recommended to be **zero** in the plains X, Y, and Z. If not, it will be generated in the scene in its corresponding position

Para los elementos de esta capa, principalmente se tiene los elementos de piso, puerta, muro y esquina.

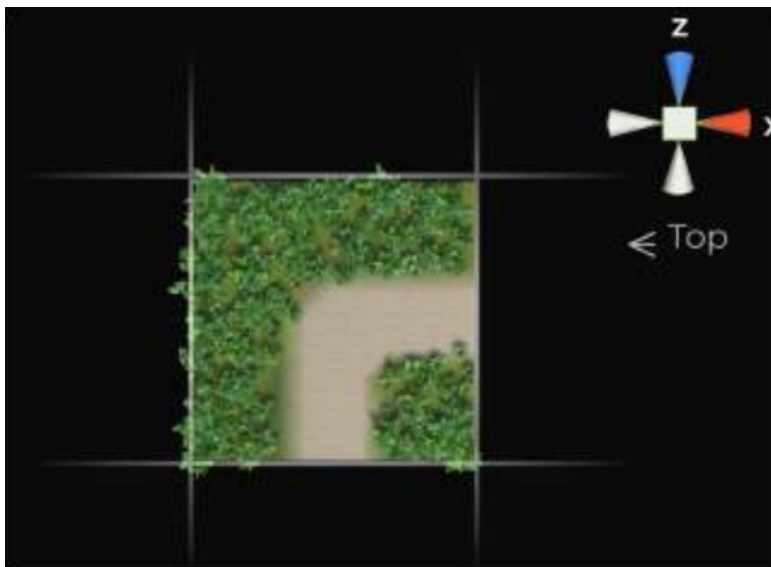
Consideraciones para los prefabs:

- La **posición** se recomienda que esté en **cero**, en los planos X, Y y Z. De no ser así, se generarán en la escena en su respectiva posición.
- Las dimensiones o tamaño de todos los elementos tienen que ser la misma. Si un mismo grupo de tile tiene dimensiones distintas, al momento de generar el nivel en la escena, habrá problemas de solapamientos o de espacios vacíos entre los tiles.
- Las direcciones y pivotes dependerá de los tipo de elementos, por ejemplo:



Exterior Layer Bundles

Esta capa trabaja principalmente con conexiones de tiles piso, para poder asignar las conexiones en los bundles es necesario saber la rotación del tile en el eje Y. En este caso, de acuerdo a la imagen la rotación en el lado Z positivo tiene una sección de pasto, y el X positivo tiene una sección de pasto con camino.



Otras consideraciones para los prefabs:

- La parte visible de los planos de piso deben estar en dirección Y positivo.
- La **posición** se recomienda que esté en **cero**, en los planos X, Y y Z, de no ser así, los tiles se generarán en la escena en su respectiva posición.
- Las dimensiones o tamaños de todos los tiles tienen que ser la misma. Si un mismo grupo de tile tiene dimensiones distintas, al momento de generar el nivel en la escena, habrá problemas de solapamientos o de espacios vacíos entre los tiles.
- El pivote está al centro del modelo.

Population Layer Bundles

La población es básicamente todo lo que sea elementos del escenario, decoración, enemigos, npc, entre otros. Para su generación, debemos considerar:

- La **posición** se recomienda que esté en **cero** en los planos X, Y y Z, de no ser así, se generarán en la escena en su respectiva posición.

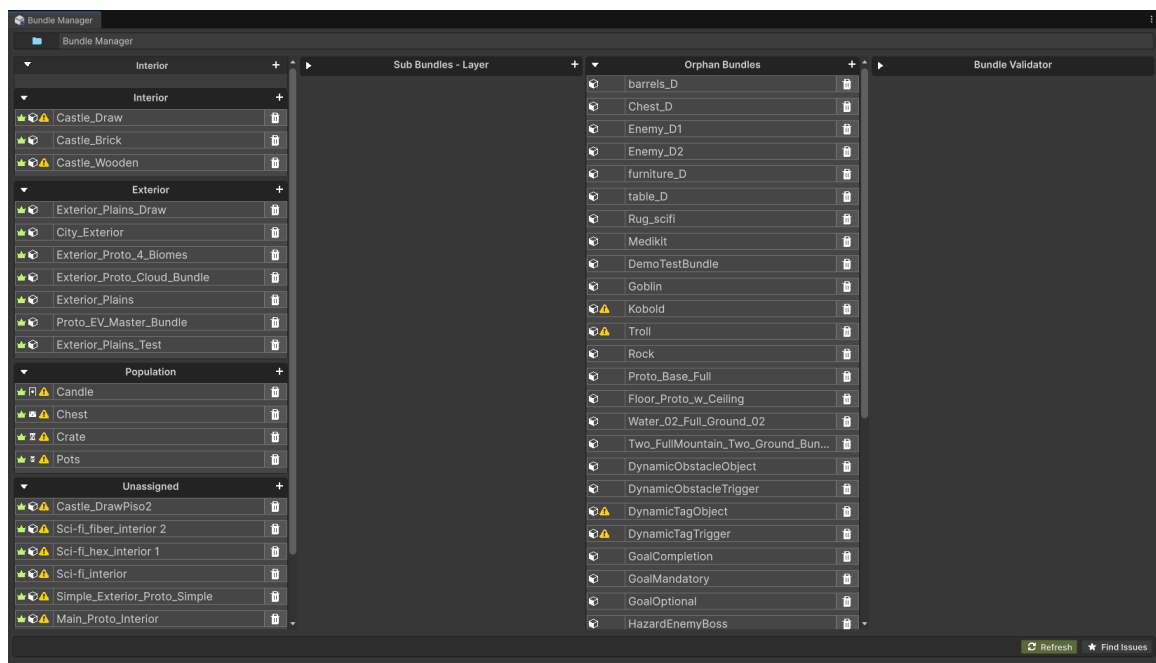
- Las dimensiones o tamaño de todos los elementos tiene que ser la misma. Si un mismo grupo de tile tiene dimensiones distintas, al momento de generar el nivel en la escena, habrá problemas de solapamiento o de espacios vacíos entre los tiles.
- Las direcciones tienen que estar todos los elementos mirando hacia un mismo plano, ya sea en el eje X o en el Z.
- Los pivotes se recomienda que todos estén en el centro inferior (0, 0, 0).



Bundle Configuration

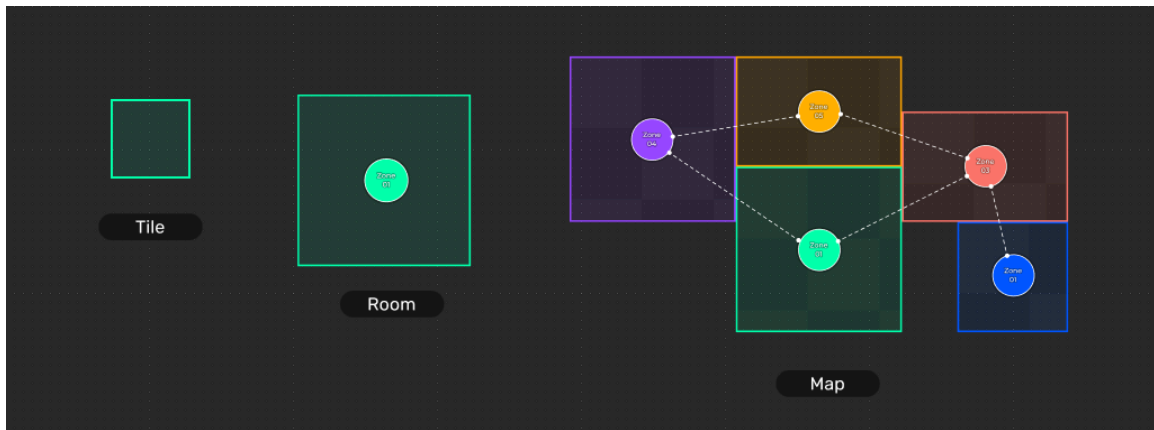
List of Bundle's Characteristics

Bundle Manager window

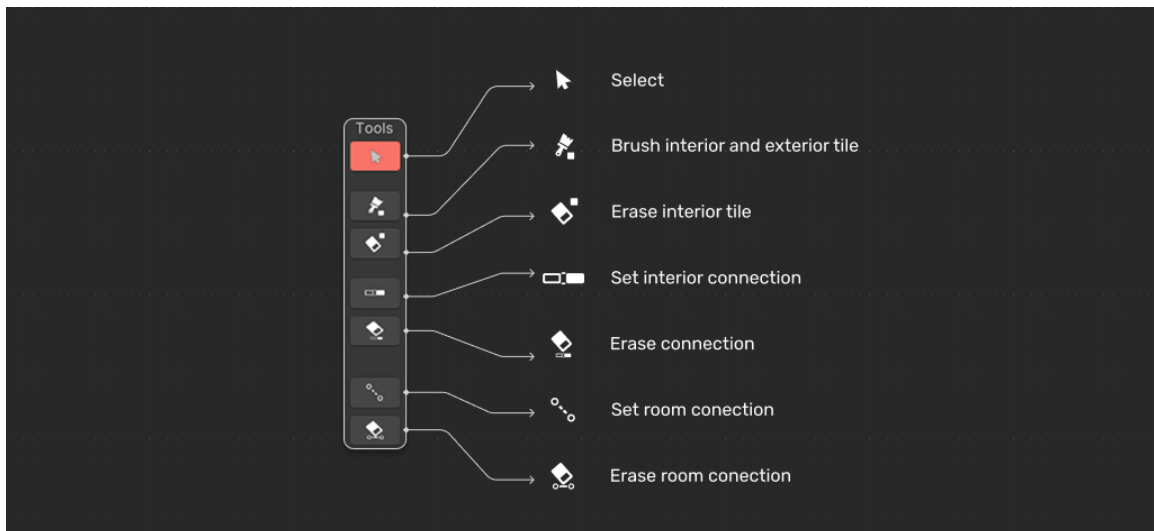


LBS - Module 1A Dungeon Generation

Module 1A is intended for designing indoor scenery, such as dungeons or buildings.



The method use a **Tilemap** approach, representing the level created in this module. The rooms can be modified by painting or removing the Tiles. You can create zones to determine the number and positions of the rooms, by changing the data of these zones you can determine their size, and by generating connections you indicate which rooms are connected by doors.



Select: With select active you can select any part of a zone to view its information. This appears in the LBS inspector option under **Current Data** section.

Brush: Allows you to paint areas or rooms.

Erase: Allows you to delete one or more tiles from the created zones.

Set interior connection: For the creation of walls, doors or placing gaps between zones.

Palette and manual usage

Assistant

The Assistant/Wizard in step 1A uses the Hill Climbing algorithm, seeking to meet 3 criteria:

- If the connection between zones is fulfilled.
- If the rooms are the right size, and any space it considers empty will be filled.
- The wizard will give more importance to the connection between zones, and will give less importance to the deletion of empty spaces within zones.

Assistant mode

After adding the connections between zones we can click on **Assistants** side panel if it is not already displayed.

A constraint panel will appear that uses artificial intelligence to know how far it can extend the dimensions of the room when creating room connections.

Optional: Press the button **Recalculate Constraints** to automatically calculate the current zone size settings if they were resized before Hill Climbing was run.

Click the **Execute** button to run the optimization algorithm, which will try to find a room configuration that meets the user's constraints.

[!Warning] When running the algorithm, the operation may take some time to complete, the time taken depends directly on the number of rooms, the number of connections and the shape of the room.

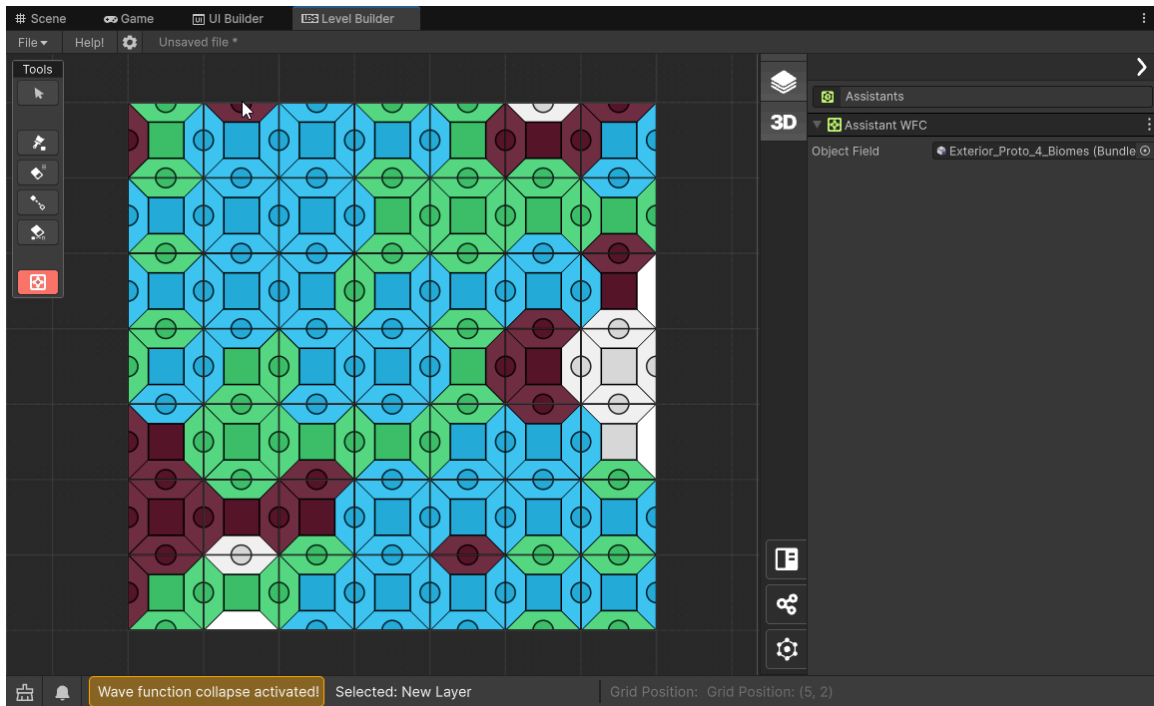
3D Generation Panel

Here you can build the current map in the 3D Scene

The LBS 3D Generator can produce mind-boggling graphics easily.

LBS - Module 1B - Wave Function Collapse

This modules layer is designed to allow creating exteriors and overlord maps.



Overview

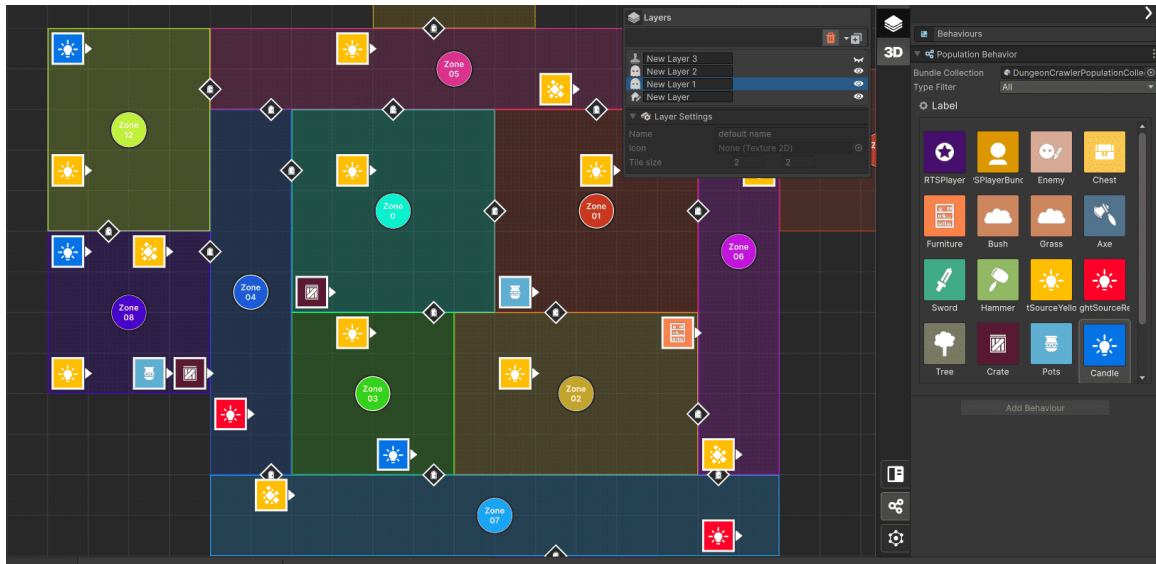
Module 1B is focused on the creation of exteriors or tiles with connections that meet organic constraints, it allows you for example to define the paths of different boxes and to define the type of connection between them. In this module, you can use the Wave Function Collapse (WFC) wizard to define suggestions for connections.

Tools

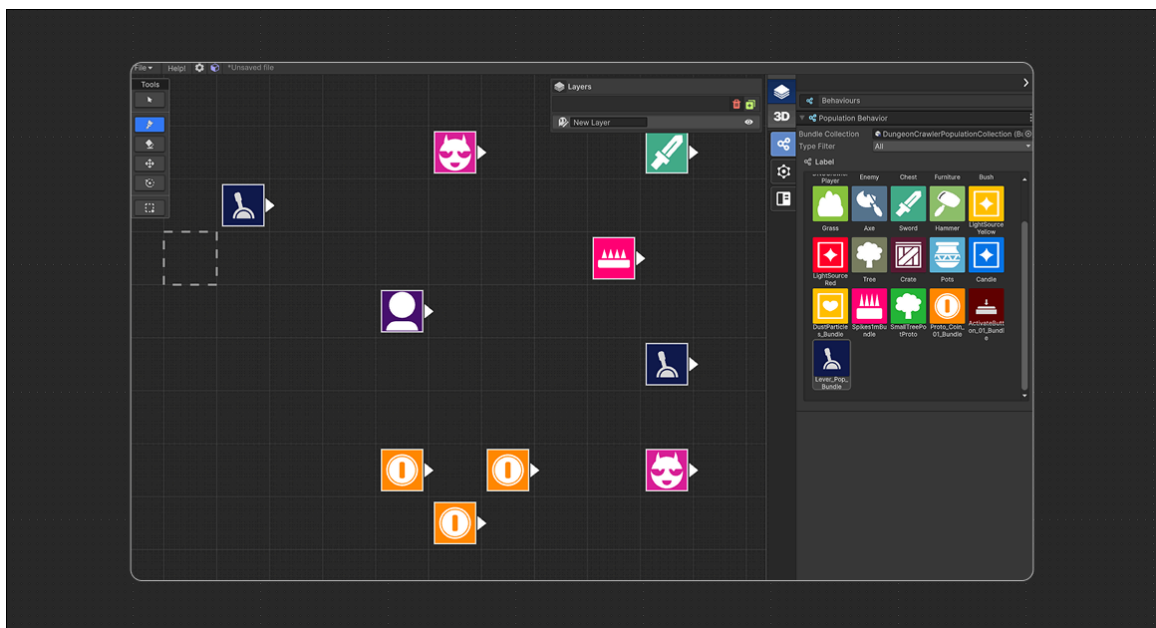
- **Select Tile:** Clicking on one of the tiles displays the coordinates of the tile in the tool inspector.
- **Brush Exterior Tile:** Paint tiles on the exterior type layer.
- **Delete tile:** Delete tile or group of tiles.
- **Paint Road tile:** Establish connection between tiles.
- **Delete Road tile:** You can delete connections by holding left click on the tile(s) that have connections, you must drag the mouse to delete.
- **Paint with Assistant:** With this wizard you can fill the selected area with connection suggestions using the algorithm. WFC is a constraint satisfaction algorithm.

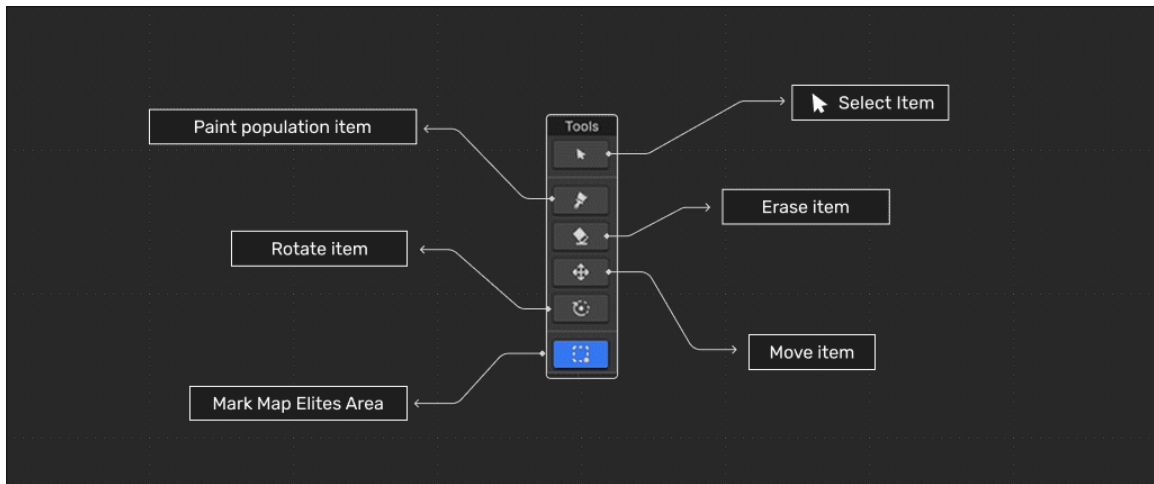
Workflow

LBS - Module 2 - Asset Population System



Module 2 - Population: describes how the population works. In the maps created, whether exterior or interior, you can place **assets**, **entities** and **decorative elements** such as enemies, players, furniture, weapons, rewards, among others. There is also an assistant called **Map-Elites**, which is an Smart Tool that orders and generates different varieties of position of the elements on the map depending on the game presets.



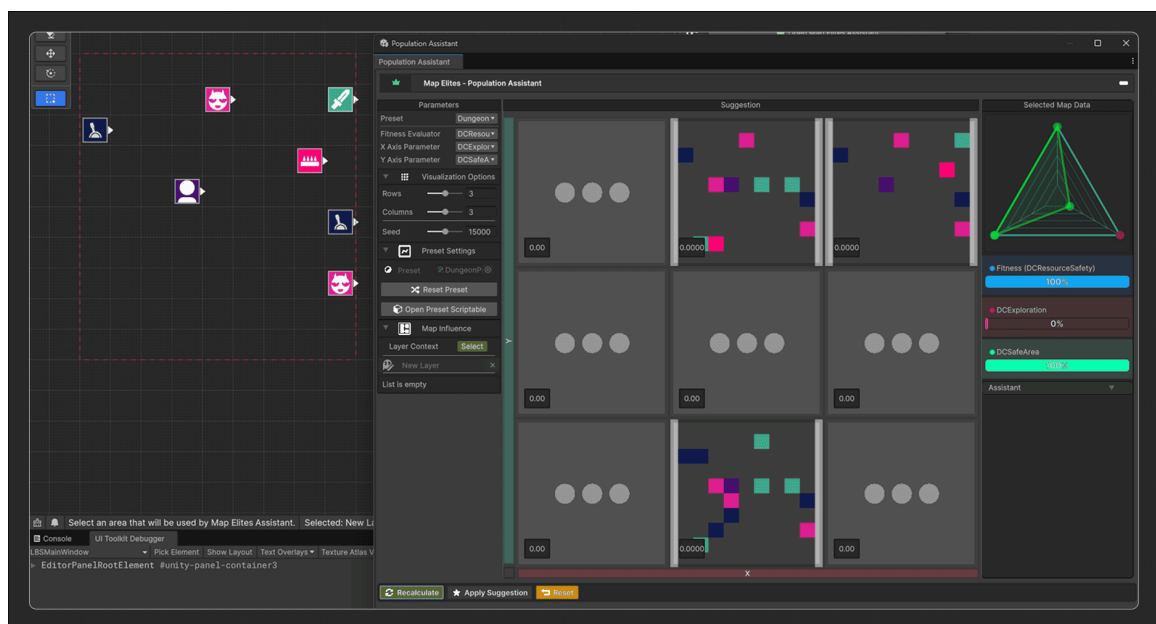


Toolset

- **Select tool:** Select a specific item and inspect it.
- **Paint Tool:** Paint with the current item selected from the LBS inspector.
- **Erase item:** Erase a item.
- **Rotate item:** Rotate a population item.
- **Move tool:** Move a current selected item.
- **Mark Map Elites Area:** Select an area to evaluate in Map Elites Windows.

Map Elites Windows

MAP-Elites is a classification algorithm that works in conjunction with an optimization algorithm to add diversity to the final result of the population map. The tool works with a two-dimensional map, X and Y, which will be the evaluators. In each iteration of the algorithm optimization, the result is located in the corresponding position on the map according to the score obtained. This score is based on the evaluators chosen by the assistant, with the lowest score being at the bottom left and the highest evaluation score being at the top right. The two evaluators (x and y) can be modified by the user and set as desired. The level of granularity depends on the number of partitions chosen.

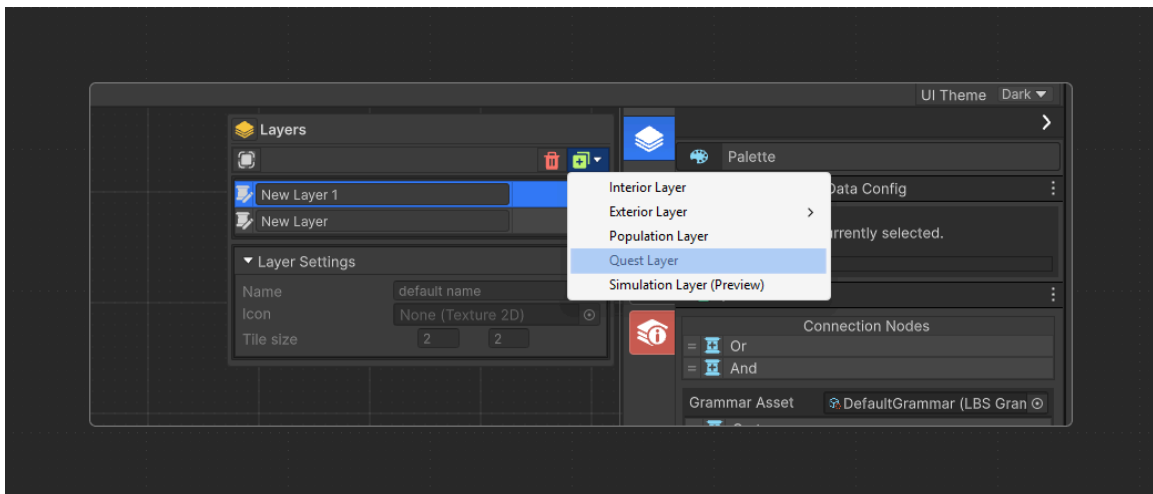


LBS - Module 3 - Quests

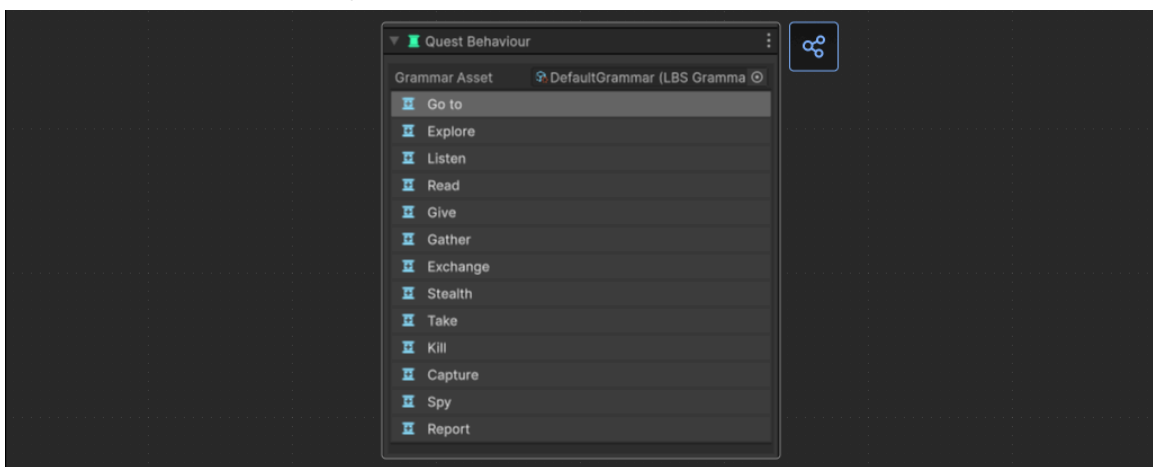
Module 3 allows you to create missions consistent with the elements in the level and a predefined set of rules or grammar. It also generates an observer and triggers in the Unity scene to manage the mission flow.

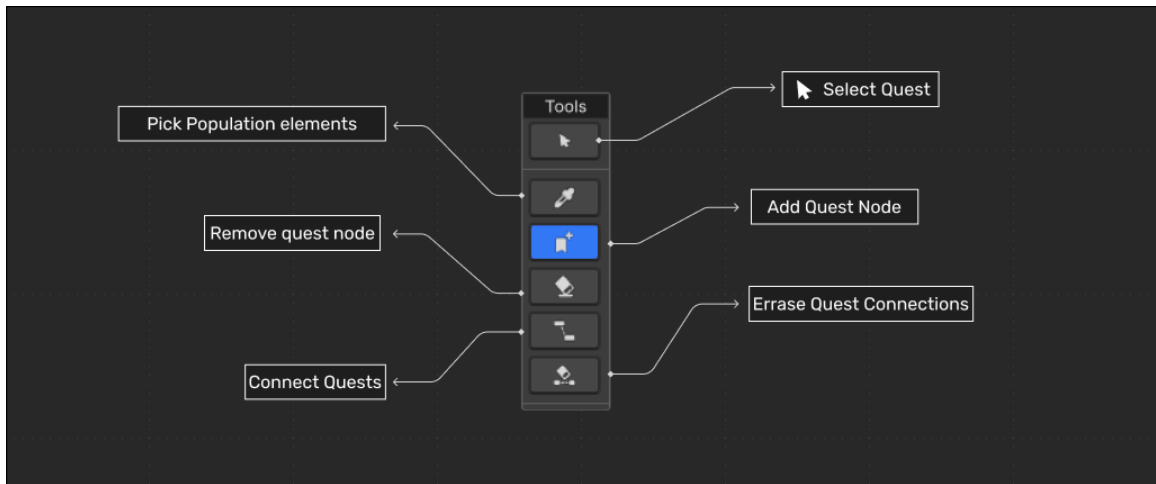
This module allow the designer to:

- Configure a grammar for the mission.
- Procedurally expand the mission tree.
- Suggest action nodes.
- Select objectives/objective types for each action.
- Validate whether the map has the necessary elements to complete the mission.
- Modify a semantic file for the grammar.



Create a new layer for quests in the bar between the LBS inspector and the workspace. Pressing it will open a layer manager window. Quest layers work the same as the other three types of windows.





Tool set overview:

- **Select Tool:**

Glossary

Action: In the quest layer, an action is a container of different grammars. Actions can be configurable in the LBS inspector

Layer: To start working we need to create a new layer, each module works in its own layer, and multiple layers can be created in the same project.

LBS Inspector: The right side panel allows the user to pick elements and tweak tools parameters.

Workspace: The space where the user can paint with layer data depending on the current layer type.